ACADEMY

Empowering For Professional Excellence

Project Report

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Course: Data Science

PITP – Batch 5, Phase 2 ,Group 2

Project Title: Student's Phone Addiction affects on Productivity.

1. INTRODUCTION:

The objective of this project is to analyze how smartphone usage and addiction affect students' productivity. Using a real-world dataset, we aim to identify key factors influencing productivity and build machine learning models to predict the productivity score.

2. Data Collection

- Dataset: Student Phone Addiction & Productivity Dataset
- Total Records: 10,000
- Features: 13 columns (numerical + categorical)

Features:

- Daily Screen Time
- Study Hours
- Sleep Hours
- Social Media Usage
- Focus Score
- Addiction Level (categorical)
- Age
- notification per day
- Coffee per day

Target Variable:

- Productivity Score (target)

3. Data Preprocessing

Data preprocessing was performed to clean and prepare the dataset:

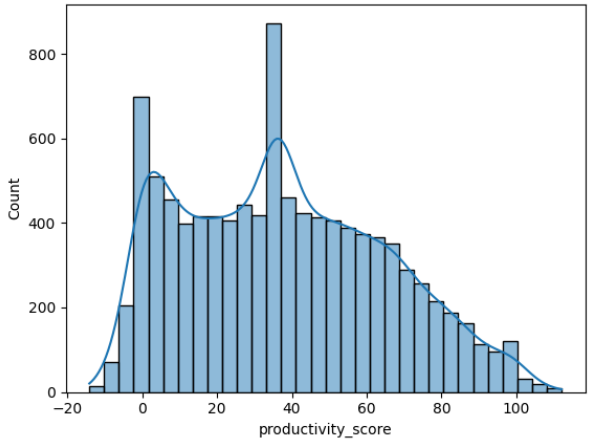
Steps:

- **Check missing values**
- **Handled Missing Values:**
  - Numerical = filled using median
  - Categorical = filled using mode
- **Encoding:**
  - addiction_level converted into numeric using Label Encoding
- **Feature Removal:**
  - Dropped: age, coffee_per_day, notifications_per_day
- **Outlier Handling:**
  - focus_score capped at 100
- **Feature Scaling:**
  - Applied StandardScaler for KNN & Linear Regression

4.Exploratory Data Analysis (EDA)

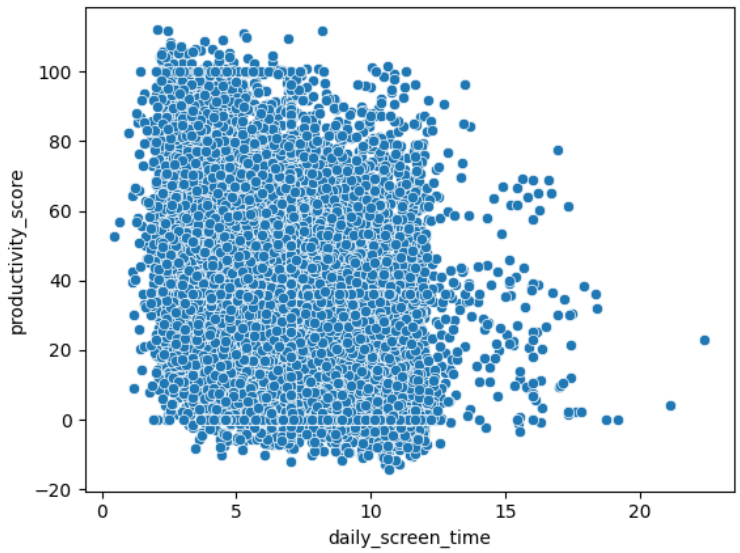
EDA is used to understand data patterns, relationships, and issues before building models

Key Visualizations:



Insights:

The productivity score follows an approximately normal distribution with most students in the mid-range. Few students have very low or very high productivity.



Insights:

As daily screen time increases, productivity decreases

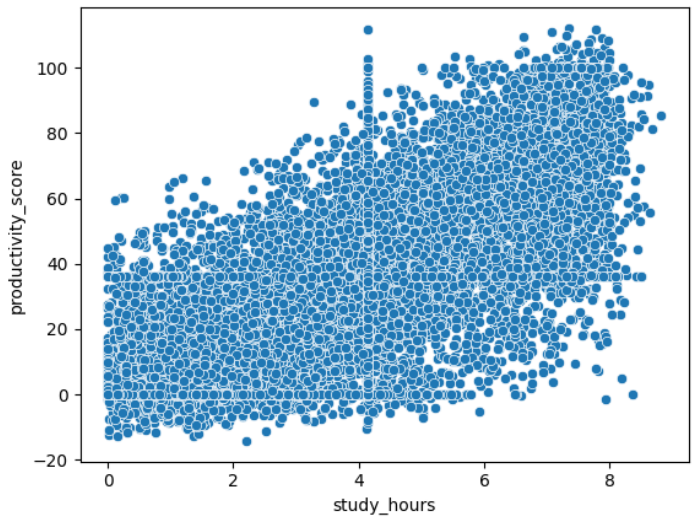
Some students:

- High screen time but still high productivity
- Low screen time but low productivity

Most data is between:

- Screen time: **3–10 hours**
- Productivity: **20–80**

This shows that phone usage impacts productivity, but it is not the only factor.

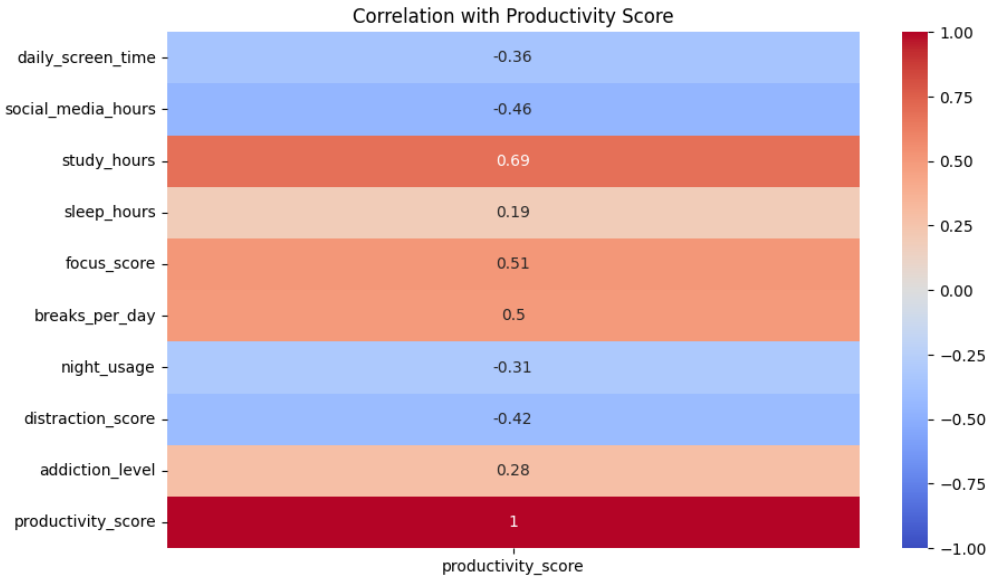


Insights:

- Clear positive relationship
- As study hours increase,productivity increases

BUT:

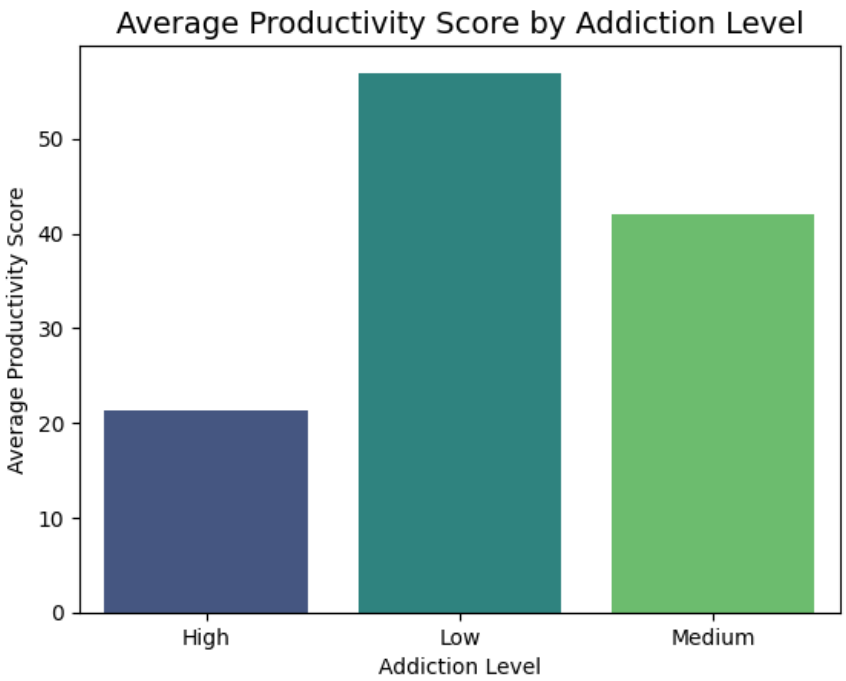
- Some students study more but still low productivity, other factors might affect them



Insight:

- Study hours = 0.68
- Focus_score = 0.51

study habits and focus have the strongest positive correlation with productivity, while social media usage and distraction negatively impact it.



Insights:

Students with high addiction levels have significantly lower productivity compared to low-addiction students.

5. Feature Engineering

New features help models learn better patterns

1. Study-to-Screen Ratio

study_hours/daily_screen_time

How much a student studies compared to how much they use their phone

Logic:

- If a student:
  - studies more

- uses phone less
  ratio will be high
- If a student:
  - studies less
  - uses phone more
    ratio will be low

2. **Lifestyle Score**

$$(0.35\times study)+(0.30\times sleep)+(0.25\times focus)-(0.30\times social_media)$$

To create **ONE combined feature** that represents:

Overall student lifestyle quality

Logic:

- **study_hours (0.35)**
  Most important = highest weight
- **sleep_hours (0.30)**
  Good sleep improves brain performance
- **focus_score (0.25)**
  Higher focus = better work

6. Model Building

Four regression models were implemented:

1. Random Forest Regressor:

- R^2 Score: **0.76**
- MAE: **10.09**
- Best performing model

2. Decision Tree

- R^2 Score: **0.72**
- MAE: **10.89**

3. K-Nearest Neighbors (KNN)

- R^2 Score: **0.73**

- MAE: **10.82**

4. Linear Regression

- R^2 Score: **0.74**
- MAE: **10.74**

7. Model Evaluation

Metrics Used:

- **MAE (Mean Absolute Error):** Average prediction error
- **R^2 Score:** Model accuracy

Comparison:

Model	MAE	R^2 Score
Random Forest	10.09	**0.76**
Decision Tree	10.89	0.72
KNN	10.82	0.73
Linear Regression	10.74	0.74

Best Model: Random Forest (highest accuracy, lowest error)

8. Tools & Technologies

- Python
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn
- Google Colab

Conclusion

- Productivity is not controlled by one factor
- It depends on a combination of habits (study, screen time, focus, lifestyle)